

TABLE OF CONTENTS

S-350 GROUT WORKS

	Page No.
1. SCOPE OF WORK.....	S.350-2
2. CODES AND STANDARDS	S.350-3
3. MATERIAL	S.350-3
4. STORAGE	S.350-5
5. PREPARATION AND MIXING	S.350-5
5.1. Surface Preparation	S.350-5
5.2. Shimming And Wedging	S.350-6
5.3. Formwork	S.350-7
5.4. Mixing	S.350-7
6. PACING AND CURING	S.350-8
6.1. General Placing Requirements.....	S.350-8
6.2. Hot Weather Placing Of Grout	S.350-8
6.3. Additional Placing Requirement For Drypack Grout	S.350-8
6.4. Additional Placing Requirements For Nonshrink Grout	S.350-9
6.5. Curing	S.350-9
7. INSPECTION AND TESTING.....	S.350-10
7.1. General Provisions	S.350-10
7.2. Testing	S.350-10
8. QUALITY ASSURANCE/QUALITY CONTROL (QA/QC).....	S.350-11

S.350 GROUT WORK

1. SCOPE OF WORK

- 1.1 The scope of grout work shall include, but not limited to, the following Bidder's work as indicated in Part 4, Subsection 1.2 "Summary of Work" and S-001 "General Design Requirement of Structure, Civil and Building Works" and shall include furnishing, mixing, placing, curing, inspecting and testing of grout as required for the work :
- Column bases and stair bases.
 - Bases and supports of all mechanical and electrical equipment.
 - All anchor bolts including sleeves.
 - All conduits, piping etc. passing through concrete floors or walls.
- 1.2 Grouting of equipment shall conform in every respect to the requirements of equipment manufacturers.
- 1.3 Installation of grout shall be in strict accordance with the manufacturer's printed instructions.
- 1.4 Wherever requirements for grout and grout installation differ between this Specification and equipment manufacturer's requirements, equipment manufacturer's requirements shall be applied, subject to review by the Owner.
- 1.5 Types and Specific Uses of Grout: shall conform to the following Table S-350-1 or approved equivalent:

TABLE S-350-1
Type and Specific Uses of Grout

Type of Grout	Specific Use (Grouting)
a. Plastic Grout (regular sand and cement grout)	a.1 Anchor bolts including sleeves and column base plates. a.2 Conduit, piping etc. through concrete floors or walls.
b. Drypack Grout	Equipment bases where required by manufacturer
c. General Grout Work Nonshrink Grout	Bases and support of mechanical and electrical equipment imposed to medium repetitive dynamic load or force and as directed by the owner.
d. Precision Gouting Nonshrink Grout	d.1 Anchor bolts in drilled holes in concrete. d.2 Equipment bases where required by manufacturer or subject to high repetitive dynamic loads.

2. CODES AND STANDARDS

- 2.1. The Bidder shall comply with the requirements of Part 4, Subsection 2.5 “Codes and Standards”.
- 2.2. The codes and standards of the following organizations extracted from the Part 4, Subsection 2.5 “Codes and Standards” are specifically applicable to the design and testing of the work included in this Specification:

ACI	American Concrete Institute
ASTM	American Society for Testing and Materials.
CRD	Chief of Research and Development, U.S. Army Corps of Engineers
JIS	Japanese Industrial Standards

3. MATERIAL

- 3.1. Grout Component and Products: shall conform to the requirements specified in Table S-350-2, unless otherwise indicated.

TABLE S-350-2
GROUT COMPONENTS AND PRODUCTS

Type of Grout	1 Cement	2 Agregated	3 Admixture (if used)	4 Mixing Water	5 Grout Product
a. Plastic Grout	a.1 Portland Cement, ASTM C150 Type II, unless otherwise indicated.	a.2.1 Grout 75 mm thick or less: Fine Aggregate, ASTM C33 a.2.2 Grout thicker than 75 mm, Fine Aggregate, ASTM C##, or Coarse Aggregate, Size No.7 or 8, ASTM C33.	Conforming to ASTM C494, Type A, Water Reducing Admixtures: containing chloride not more than 1% by weight. (use only if reviewed by the Owner).	a.4 Clean, potable, and free from injurious amounts of oil, acid ,salt, alkali, organic matter and other deleterious substances.	Not specified
b. Dry-pack Grout	Same as Item a.l.	Fine Aggregate, ASTM C33	---	Same as Item a4.	Not specified
c. General Grout Work Non Shrink	Manufacturer's standards (This grout shall not contain oxidizing and gas producing agents, catalysts, inorganic accelerators and chlorides) ASTM-C109			Same as Item a4	Not specified
d. Precision Grout	Manufacturer's standards (This grout shall not contain any metallic aggregates, oxidizing and gas producing agents, catalysts, and more than 300 ppm chloride.) ASTM – C109			Same as Item a4	Not specified

- 3.2. Grout Properties: shall conform to the requirements of Table S-350-3, unless otherwise indicated.

TABLE S-350-3
Property Requirement for Placed Grout

Type of Grout	Minimum Compressive Strength	Consistency	Expansion	Shrinkage	Bleeding
a. Plastic Grout	250 kg/cm ² at 28 days, unless otherwise specified.	Plastic with maximum slump 125 mm for grout 75 mm thick or less and maximum slump 75 mm for grout thicker than 75 mm	No requirement	No requirement	No visible bleeding
b. Dry Pack Grout	350 kg/cm ² at 28 days, unless otherwise specified.	Stiff with hand shaping workability, and no measurable slump.	No expansion	Not more than 1.0 %	No bleeding
c. General Grout Nonshrink	700 kg/cm ² at 28 days, ASTM – C109	Flowable, 35-45 cm lateral flow at 22 ^o – 30 ^o C.	Not more than 0.4% at 7,14 or 28 days CRD 621	No shrinkage	No bleeding
d. Precision Grout Non shrink	800 kg/cm ² at 28 days, ASTM – C109	Flowable, 45-55 cm at 22 ^o – 30 ^o C with time of efflux not more than 30 second. ASTM 1107	Not more than 0.4% at 7,14 or 28 days CRD621	No shrinkage	No bleeding

4. STORAGE

All materials shall be stored in such a manner as to avoid contamination, evaporation or damage. Cement shall be stored in weather tight buildings, bins or silos which will exclude moisture and contaminants, liquid admixtures shall be protected from temperature changes which would adversely affect their characteristics.

5. PREPARATION AND MIXING

5.1. SURFACE PREPARATION

- 5.1.1 Concrete surfaces to be grouted shall first be cleaned and then be uniformly roughened to a minimum depth of 3 mm. Prior to placing grout, all concrete

or metal surfaces to be in contact with grout shall be thoroughly cleaned to remove all laitance, oil grease, paint, dirt, rust, loose mill scale, damaged concrete and any other loose particles that may interfere with complete bearing and bond or may react with grout materials. Special care shall be taken to assure that all waste materials and water are removed from anchor bolt holes and sleeves in concrete.

- 5.1.2 The cleaned concrete surfaces to be in contact with plastic grout shall be dampened but not saturated and free surface moisture shall be removed just prior to grout placement.
- 5.1.3 For drypack grout, the equipment manufacturer's requirements for surface preparation shall be met, but as a minimum the followings are required:
 - 5.1.3.1 The cleaned surfaces of the concrete to be in contact with drypack grout shall be presoaked for at least 6 hours before grouting.
 - 5.1.3.2 Just prior to placing grout, the concrete surfaces should be blown with compressed air to remove any free water on these surfaces.
- 5.1.4 All cleaned surfaces to be in contact with nonshrink grout shall also be prepared in accordance with manufacturer's instructions.
- 5.1.5 For metal surfaces where bond between metal surfaces and grout is not desired flake graphite, paraffin or form release agent shall be applied to these surfaces, and no grease or oil shall be used.

5.2. SHIMMING AND WEDGING

- 5.2.1 Prior to placing grout, column base plates and other structural elements shall be properly supported with shims or wedges, the smaller dimensions of which shall be oriented in the direction of grout flow or packing. Mechanical and electrical equipment shall be shimmed or wedged before grouting in strict accordance with equipment manufacturer's printed instructions. Unless otherwise specified, shims and wedges to be removed shall be made of carbon or stainless steel, and those to be left in place shall be made of stainless steel.
- 5.2.2 All shims and wedges used for support shall be carefully removed except those required by the equipment manufacturers to be left in place, after the

grout has been set and hardened. The voids resulting from the removal of these shim, wedges and leveling screws, shall be filled with the same grout unless otherwise specified. All shims and wedges shall be coated with a film of paraffin or form release agent to facilitate their removal, and no grease or oil shall be used.

5.3. FORMWORK

Formwork for placing grout shall conform to grout or equipment manufacturer's recommendations and the applicable requirements of Chapter 4 of ACI 301.

5.4. MIXING

Grout shall be thoroughly and uniformly mixed in proper proportions in accordance with the following requirements and/or the grout manufacturer's instructions, in order to obtain the properties indicated in Table S-350-3.

- 5.4.1 Batches of plastic grout for mixing shall be less than 0.2 m³ and shall be mixed in a paddle type mortar mixer for not less than 1 minute, with the water added slowly to assure to efficient mixing and water cement ratio not exceeding 0.5 by weight.
- 5.4.2 For drypack grout, the equipment manufacturer's requirements for mixing shall be met, but as minimum the followings are required:
 - 5.4.2.1 Drypack grout shall be thoroughly and uniformly mixed in a paddle-type mortar mixer for at least 5 minutes, and in quantities that can be placed within one hour.
 - 5.4.2.2 Grout mix proportion by weight shall be 1 part of cement to 2-1/4 parts of fine aggregates, with water content that will give a specified slump as shown table S 350 – 2 or ASTM C143 and no excluded water when packing.
 - 5.4.2.3 Grout shall be remixed every 30 minutes; however, retempering of grout will not be permitted.
- 5.4.3 Nonshrink grout shall be mixed in strict accordance with manufacturer's instructions.

6. PLACING AND CURING

6.1. GENERAL PLACING REQUIREMENTS

Grout shall be placed in strict accordance with the following requirements and/or manufacturer's grouting procedures.

6.1.1 Grouting of a plate or base shall be a continuous operation until the whole plate or base is completely grouted, except that shimmed and wedged areas shall be grouted in accordance with Clause 5.2.

6.1.2 Special care shall be taken to ensure that space under grouted base plate or equipment base is entirely filled with grout. For placing plastic grout or non-shrink grout, grout shall be poured from one side and continuously rodded with care until it flushes up to the top of the form on all other sides and through grout holes, if any, of the plate/base. Base plate with shear bars in more than one direction may be grouted through a central hole, only if air-relief holes are provided in the vicinity of the shear bars where air may be trapped.

6.2. HOT WEATHER PLACING OF GROUT

6.2.1 Grout temperature during placing shall not exceed 32°C, unless otherwise indicated.

6.2.2 Retarding admixtures shall not be used unless otherwise indicated.

6.3. ADDITIONAL PLACING REQUIREMENT FOR DRYPACK GROUT

Dry pack grout placement shall meet the equipment manufacturer's requirements, but as a minimum the followings are required:

6.3.1 Grout shall be placed and packed within one hour after mixing.

6.3.2 Grout shall be placed and packed in small amounts to eliminate the possibility of a porous pack. Each layer shall be thoroughly rammed into place before succeeding layers are placed, and this procedure shall be repeated until the form is packed full.

6.4. ADDITIONAL PLACING REQUIREMENTS FOR NONSHRINK GROUT

Non-shrink grout shall be placed in accordance with manufacturer's detailed instructions.

6.5. CURING

6.5.1 Curing of Plastic Grout

6.5.1.1 Exposed surfaces of grout shall be cured immediately after completion of grout finishing and/or form removal by applying curing compound.

6.5.1.2 Curing compound shall conform to ASTM C309 and shall be applied in accordance with manufacturer's recommendations, immediately after any water sheen which may develop after finishing has disappeared from the grout surface.

6.5.1.3 Exposed surfaces of grout and the plate in contact with grout shall be protected from heating by the sun for at least a week after grout placement.

6.5.1.4 No load shall be applied on the grout until compressive strength of the grout reaches 75 percent of the specified strength or until the grout has been moist cured for at least 7 days.

6.5.2 Curing of drypack grout shall meet the equipment manufacturer's requirements, but as a minimum the followings are required:

6.5.2.1 Exposed surfaces of grout shall be kept continuously moist at least 7 days after grout placing. Following moist curing and while the surface is still damp, a coat of curing compound shall be applied.

6.5.2.2 Moist curing may be achieved by any on the following methods:

- a. Ponding or continuous sprinkling
- b. Application of absorptive mats or fabric continuously kept wet
- c. Application of sand continuously kept wet

6.5.2.3 Exposed surfaces of grout and the plate in contact with grout shall be protected from heating by the sun for at least a week after grout placement.

6.5.3 Curing of Non-shrink Grout

Non-shrink grout shall be cured in accordance with manufacturer's instructions.

7. INSPECTION AND TESTING

7.1. GENERAL PROVISIONS

7.1.1 The Bidder shall be responsible for all testing work required by this Specifications unless otherwise indicated.

7.1.2 The Owner may, during the course of the work, inspect the various phases of the work for full compliance with all requirements of this Specification.

7.2. TESTING

7.2.1 All grout for the work shall be tested in accordance with the requirements of Table S-350-4, "GROUT TESTING". The reports of the test result shall be submitted to the Owner for approval.

7.2.2 Initial tests shall be performed prior to the use of the grout. All grout components, including the water, to be used for the initial tests shall be those specified for use in the works.

TABLE S-350-4
Grout Testing

Test	ASTM Designation	Frequency
Compressive strength at 7 and 28 days	ASTM C109 or ASTM C31 ASTM C39 CRD-C261	One test for each grout mix or as directed by the Owner
Consistency	ASTM C 1107	One test for each grout mix or as directed by the Owner

8. QUALITY ASSURANCE/QUALITY CONTROL (QA/QC)

- 8.1. Quality control shall conform to the requirements of Part 4, Subsection 2.7
- 8.2. The quality control procedures and documentation so specified in shall include, but not be limited to, the following:
 - 8.2.1 Testing and placing.
 - 8.2.2 Packaging and shipping.

(Last page of S-350)